3.4.1 Wave energy transfer and vibration direction

In a mechanical wave, particles oscillate near their equilibrium positions and transfer energy to neighbouring regions rather than travelling along with the waveform. Transverse vibrations are perpendicular to propagation. Longitudinal vibrations are parallel. An up-and-down plot of longitudinal displacement does not mean the particles themselves move vertically.

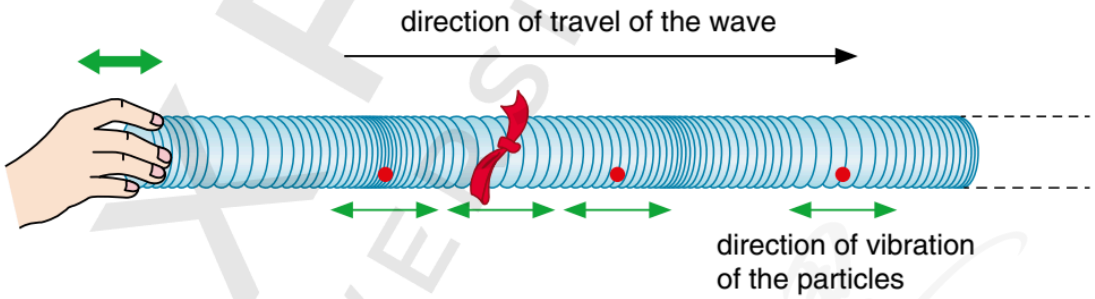


Fig 4.1i *On a longitudinal wave, particles vibrate along the direction the wave travels.*

The particle vibration arrows are parallel to wave travel. Energy propagates while particles oscillate about equilibrium positions.

3.4.2 Amplitude

Amplitude is the maximum displacement from equilibrium and has units of length. On a transverse-wave plot, measure from the equilibrium line to a crest or trough. Crest-to-trough height is twice the amplitude. Amplitude measures oscillation size, not wavelength or how far a particle travels along the propagation direction.

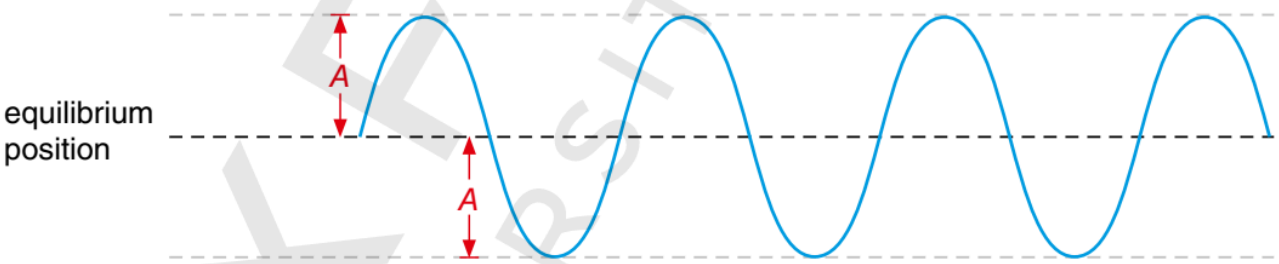


Fig 4.2a *Amplitude of a transverse wave.*

Amplitude is measured from the equilibrium line to a crest or trough; crest-to-trough distance is twice the amplitude.

3.4.3 Wavelength

Wavelength is the shortest separation along propagation between adjacent points in phase at the same instant, such as neighbouring crests, troughs or compression centres. A crest and the next trough are half a wavelength apart.

In-phase particles are at the same stage of their oscillation cycle. Compare both displacement and direction of motion. They cross equilibrium in the same direction together and reach maximum displacement on the same side together.

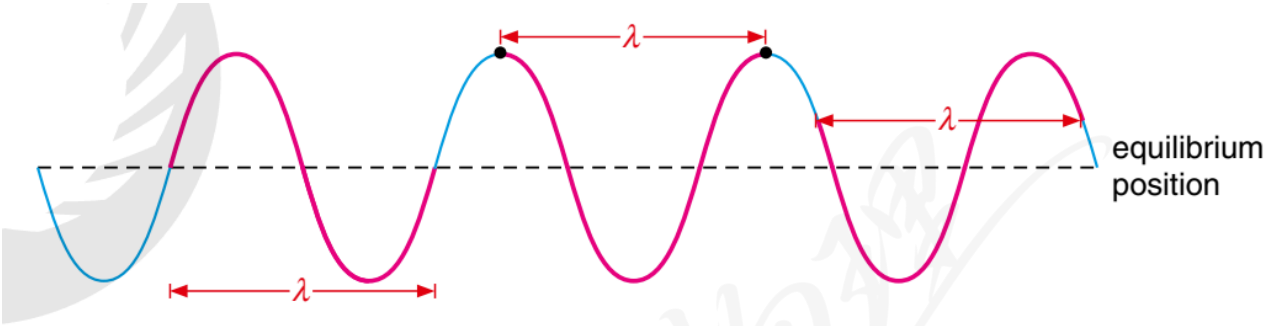


Fig 4.2b *Wavelength of a transverse wave.*

One wavelength separates adjacent points in phase, such as neighbouring crests, measured along propagation.

3.4.4 Period and frequency

Period is the time for one complete oscillation. Frequency is the number of oscillations per second. They are reciprocals, measured in s and Hz respectively. On a displacement-time graph, the time between adjacent crests gives period. The separation between crests on a distance graph instead gives wavelength.

$$f = \frac{1}{T}$$

f: frequency (Hz); T: period (s).

3.4.5 Wave speed, frequency and wavelength

Wave speed describes how fast the waveform or phase propagates. In one period it advances one wavelength, so speed equals frequency times wavelength. This differs from the instantaneous speed of a particle oscillating locally. Use consistent units and ensure all quantities refer to the same wave in the same medium.

$$v = \frac{d}{t} = \frac{\lambda}{T} = f\lambda$$

v: wave speed (m s^{-1}); d: distance travelled by the wave (m); t: time (s); λ: wavelength (m); T: period (s); f: Hz. This is not particle vibration speed.

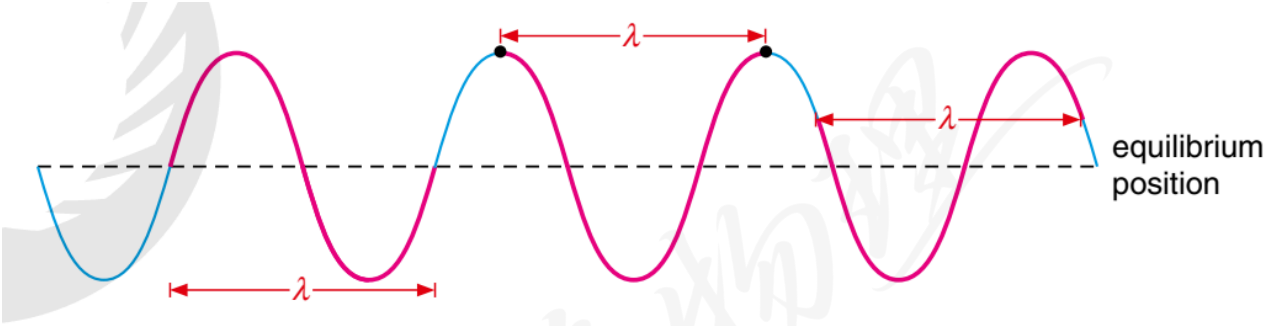


Fig 4.2b *Wavelength of a transverse wave.*

One wavelength separates adjacent points in phase, such as neighbouring crests, measured along propagation.

3.4.6 Speed of waves on a string

Waves on a taut string or spring travel faster with greater tension and slower with greater mass per unit length. Speed depends on the medium and its state, not simply on how rapidly the source is shaken. With unchanged medium conditions, increasing source frequency shortens wavelength.

3.4.7 Particle motion in a travelling wave

In an ideal sinusoidal travelling wave in a uniform medium without attenuation, particles share amplitude, frequency and period, but their phases depend on position. In-phase particles have the same displacement and motion direction. Antiphase particles have opposite displacements and velocities and are half a cycle apart.

A displacement-position graph shows different particles at one instant. A displacement-time graph follows one particle. Comparing waveforms at nearby times reveals a particle's direction of motion. A moving crest shows the wave pattern advancing; particles still oscillate around their equilibrium positions.